Kartik Bharat Sonaghela

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TECHNICAL SKILLS

- **Programming Languages**: Python, SQL
- **Databases**: MySQL, VectorDB
- **Frameworks**: PyTorch, LangChain
- **DevOps, Cloud & Misc.:** Docker, Git, Postman, Flask, CI/CD, AWS,Mlops
- **Data-Science & AI**: Machine Learning, Deep Learning, NLP, RAG, GenAI, LLMs, Fine-Tuning, AI-Agents

EXPERIENCE

Sr Data Scientist, DeepNeura Technologies May 2022 – Present| Mumbai, India

RAG-Based Chatbot for Farmers:

- Developed a RAG-based multilingual chatbot to assist farmers by answering agronomy-related queries.
- Integrated LangChain framework, MongoDB vector database, and gpt-4o-mini to enable real-time, data-backed responses.
- Enhanced farmer engagement and decision-making by delivering personalized, Indian-language support using Bayer's internal knowledge base.

Crop Trend Prediction:

- Built a crop price prediction system using Agmarknet agricultural data to help Bayer's team gain actionable mandi insights.
- Cleaned and pre-processed large datasets, engineered key features, and applied the XGBoost algorithm with parameter tuning for optimal results.
- Achieved 80% prediction accuracy across multiple mandis, enabling reliable 10-day price forecasts and supporting data-driven decisions for Bayer's agronomy and sales teams.

Jr Data Scientist, Care Risk Solution March 2021- April 2022|Mumbai, India

- Developed an Auto-ml Software which include entire machine learning cycle life starting from uploading data to predicting the data.
- Text Summarization of scraped news article into 200 words.
- Created a covid dashboard using Tableau during the second wave which shows information regarding vaccination, how many are positive and how many have recovered.

PROJECTS

YouTube Comment Sentiment Analysis Chrome Extension

- Built a Chrome extension to analyze YouTube comment sentiment in real time for audience feedback analysis.
- Trained an XGBoost classifier using TF-IDF, SMOTE, and Optuna; tracked experiments with MLflow and managed datasets with DVC.
- Deployed via Flask APIs in Docker on AWS, achieving an F1 score of 0.82, and integrated with a frontend built in HTML, CSS, and JavaScript.

Fine-Tuned Llama for Indian laws

- Fine-tuned a 3.18B LLaMA 3.1 model on 35K legal records including BNS sections and public laws to simplify legal understanding.
- Created instruction-tuned datasets with help from Gemma 2.0 flash for clearer legal explanations.
- Enabled fast, accurate legal Q&A and document understanding (e.g., FIRs) using vLLM for inference on L4 after fine-tuning on A100.

AI Code Assistant

- Built a VS Code extension that generates Python code from instructions using a fine-tuned Phi-3 4-bit quantized model.
- Used LoRA with the Unsloth library to fine-tune on iamtarun/python_code_instructions_18k_alpaca for efficient local execution.
- Delivered fast, accurate code completions within VS Code, enhancing developer productivity without relying on cloud apis.

EDUCATION

University of Mumbai **Aug 2016 – Oct 2020**
Bachelor of Engineering in Computer Engineering Mumbai, India